

Cosmologix: Fast, accurate and differentiable distances in the universe with JAX

Marc Betoule ¹¶, Jérémy Neveu ², Dylan Kuhn ¹, Maude Le Jeune ³,
Mathieu Bernard ¹, Nicolas Regnault ¹, and Sébastien Bongard ¹

¹ LPNHE, CNRS, France  ² IJCLab, Orsay, France  ³ APC, Paris, France  ¶ Corresponding author

DOI: [10.21105/joss.09633](https://doi.org/10.21105/joss.09633)

Software

- [Review](#) 
- [Repository](#) 
- [Archive](#) 

Editor: [Tobias Buck](#)  

Reviewers:

- [@cgiovanetti](#)
- [@dpiras](#)

Submitted: 26 May 2025

Published: 20 May 2026

License

Authors of papers retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC BY 4.0](#)).

Summary

Type-Ia supernovae serve as standardizable candles to measure luminosity distances in the universe. *Cosmologix* accelerates and simplifies cosmological parameter inference from large datasets by providing fully differentiable calculations of the distance-redshift relation as a function of cosmological parameters. This is achieved through the use of JAX ([Bradbury et al., 2018](#)), a Python library providing automatic differentiation and compilation for CPU and hardware accelerators. *Cosmologix* incorporates the density evolution of all relevant species, including neutrinos. It also provides common fitting formulae for the acoustic scale so that the resulting code can be used for fast cosmological inference from supernovae in combination with BAO or CMB distance measurements. We checked the accuracy of our computation against CAMB, CCL and *astropy.cosmology*. We demonstrated that our implementation is approximately ten times faster than existing cosmological distance computation libraries, computing distances for 1000 redshifts in approximately 500 microseconds on a standard laptop CPU, while maintaining an accuracy of 10^{-4} mag in the distance modulus over the redshift range $0.01 < z < 1000$.

Statement of need

Many software packages are available to compute cosmological distances including *astropy* ([Astropy Collaboration, 2013](#)), CAMB ([Challinor & Lewis, 2011](#)), CLASS ([Lesgourgues, 2011](#)) or CCL ([Chisari et al., 2019](#)). To our knowledge only *jax-cosmo* ([Campagne et al., 2023](#)) and *cosmoprime* ([Mattia, 2022](#)) provide automatic differentiation through the use of JAX. Unfortunately, at the time of writing, the computation in *cosmoprime* does not seem to be jittable and distance computation in *jax-cosmo* is neglecting contributions to the energy density from neutrinos and photons. The accuracy of the resulting computation is insufficient for the needs of the LEMAITRE analysis, a compilation of type-Ia Supernovae joining the very large sample of nearby events discovered by the Zwicky Transient Facility ([Rigault et al., 2025](#)) to higher redshift events from the Supernova Legacy Survey ([Astier et al., 2006](#)) and the Subaru Strategic Program ([Yasuda et al., 2019](#)). The LEMAITRE collaboration is therefore releasing its internal code for computing cosmological distances. The computation follows standard methods, but our JAX implementation is optimized for speed while maintaining sufficient accuracy.

Computations of the homogeneous background evolution

The core library offers JAX functions to compute the evolution of energy density in the universe (via the *cosmologix.densities* module) and derived quantities, such as cosmological distances

(via the `cosmologix.distances` module). Details are provided in the documentation. As an example, we highlight the speed and accuracy of calculating the distance modulus (the logarithm of luminosity distance) for a large number of redshifts in the following discussion.

Accuracy

The distance computation involves the numerical evaluation of an integral. The resolution of the quadrature used for this evaluation is adjustable in `cosmologix`. To assess the numerical accuracy of our baseline computation, we compared it to the same integral evaluated at 10-fold higher resolution. The difference is displayed in Figure 1 for the baseline Planck Λ CDM model, reported in Table 1 in Planck Collaboration et al. (2020). The difference in distance modulus between the coarse (baseline) and fine resolution computation is smaller than 10^{-4} mag over the redshift range $0.01 < z < 1000$, dominated by the interpolation error.

We also compared the results of various external codes to the fine quadrature of `cosmologix` as the reference. It demonstrates agreement within a few 10^{-5} mag over the same redshift range. Residual discrepancies between libraries stem from differences in handling the effective number of neutrino species. We adopt CAMB's convention, where all species share the same temperature, resulting in closer alignment with its predictions. We exclude `jax-cosmo` from this comparison because it does not account for neutrino contributions to energy density, precluding a meaningful comparison.

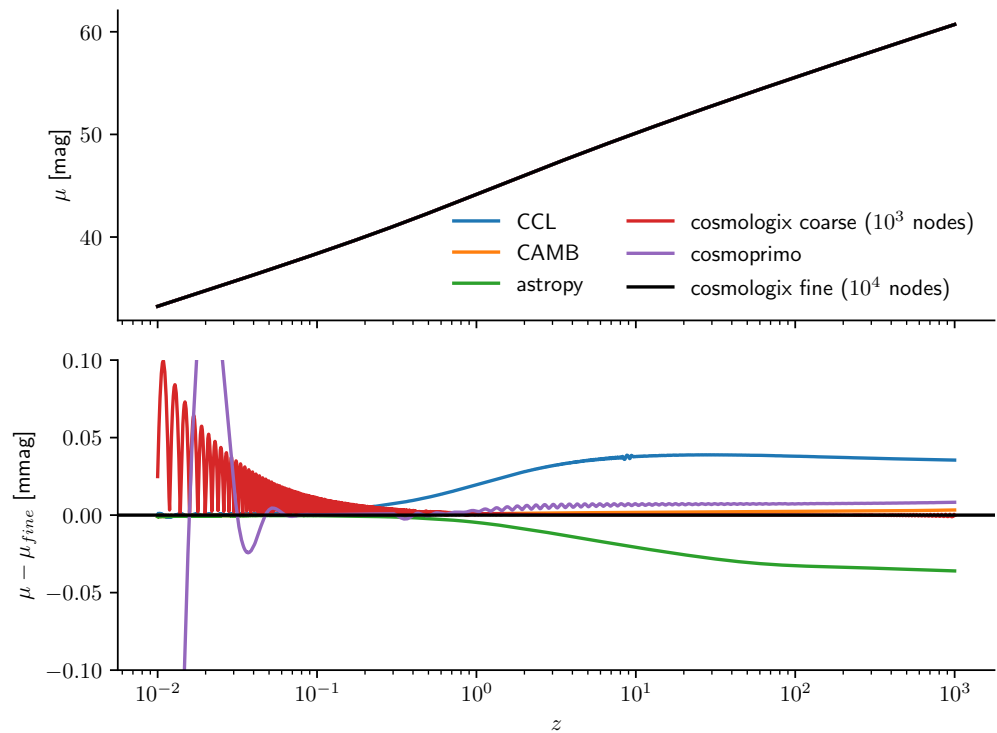


Figure 1: Top: Distance modulus for the Planck best-fit Λ CDM model as a function of redshift computed using 2 different accuracy settings in `cosmologix`, and compared to 4 other numerical libraries. Bottom: Difference in the above numerical results with respect to the higher resolution quadrature computation in `cosmologix`.

Computation speed

The computation time for a vector of distance moduli across various redshifts is plotted in Figure 2 as a function of the number of redshifts requested. We differentiate between the first

call and subsequent calls, as the initial call may involve specific overheads. For cosmologix, this includes JIT-compilation times, which introduces a significant delay. In subsequent calls, cosmologix overperforms all other tested codes by a significant margin, typically 10 times faster or more on the tested CPU architecture for 2000 redshifts (corresponding to the current number of supernovae in Hubble diagrams).

In addition we also timed the computation of the jacobian matrix ($\vec{\nabla}\mu$) of the distance modulus with respect to the 9 cosmological parameters. It is evaluated as `jax.jacfwd(mu)`. The computation time for the Jacobian is roughly 5 times larger than the function itself. This is faster than finite differences, which require 10 function evaluations, reducing computation time by approximately 50%.

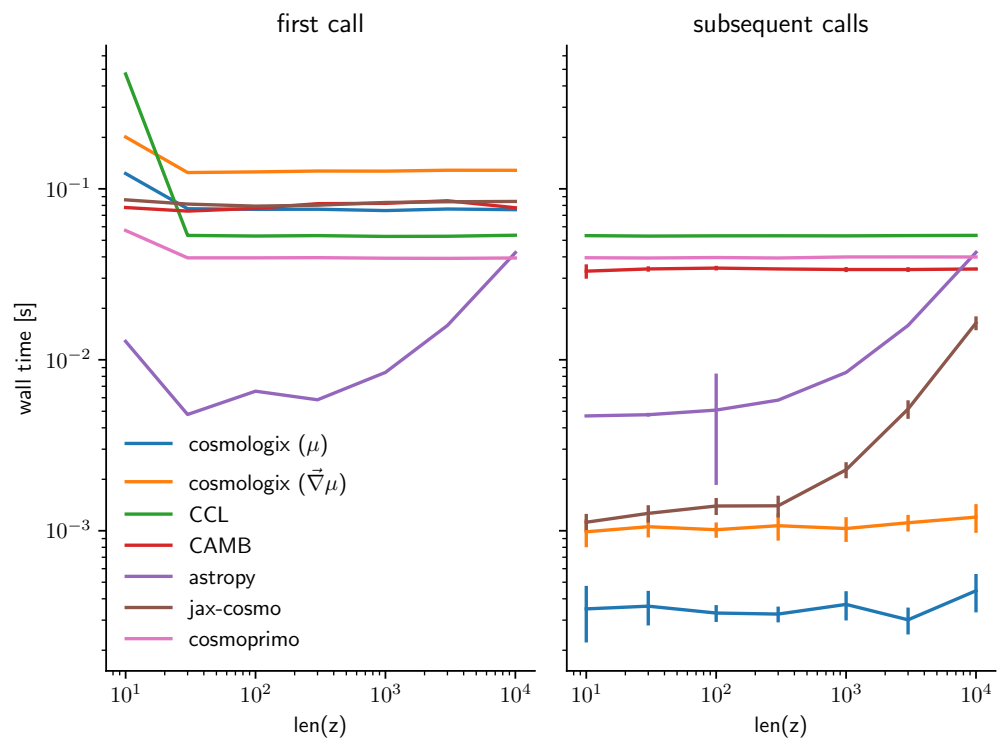


Figure 2: Computation speed of the distance modulus for various cosmological codes. The left panel displays the measured time for the first call which integrates pre-computation and in the case of JAX codes overhead associated with jit compilation. The right panel displays the median time measured over 10 subsequent calls. The error bar is the rms over 10 measurements. The measurements were obtained on an AMD RYZEN AI MAX+ 395, without GPU acceleration.

Differentiability and likelihood maximization

Last, the code provides a framework to efficiently build frequentist confidence contours for cosmological parameters for all measurements whose likelihood can be expressed as a chi-square. [Figure 3](#) provides an example 2-dimensionnal confidence region in the plane (Ω_{bc}, w) for a flat w -CDM model as probed by the Union3 supernovae compilation ([Rubin et al., 2023](#)). Ω_{bc} is the combined density parameter of baryonic and cold dark matter. The full computation took 3.86s on an Intel(R) Core(TM) i7-1165G7 at 2.80GHz without GPU acceleration.

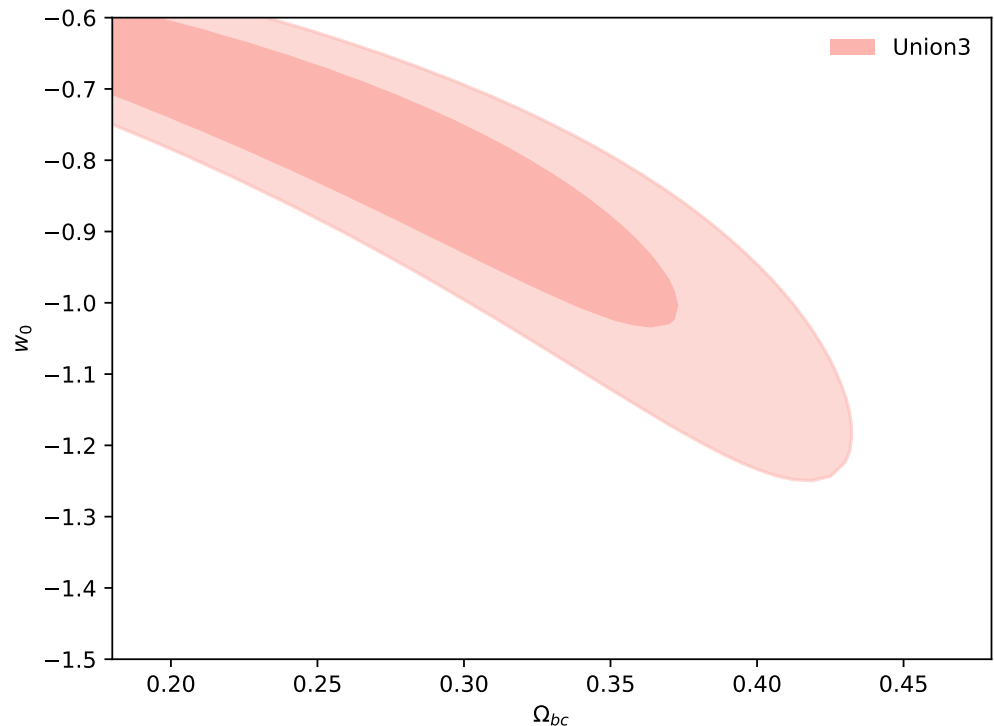


Figure 3: Confidence region at 68 and 95 percent for the w and Ω_{bc} parameters probed by the Union3 compilation.

References

- Astier, P., Guy, J., Regnault, N., Pain, R., Aubourg, E., Balam, D., Basa, S., Carlberg, R. G., Fabbro, S., Fouchez, D., Hook, I. M., Howell, D. A., Lafoux, H., Neill, J. D., Palanque-Delabrouille, N., Perrett, K., Pritchet, C. J., Rich, J., Sullivan, M., ... Walton, N. (2006). The Supernova Legacy Survey: measurement of Ω_M , Ω_Λ and w from the first year data set. *Astronomy and Astrophysics*, 447(1), 31–48. <https://doi.org/10.1051/0004-6361:20054185>
- Astropy Collaboration. (2013). Astropy: A community Python package for astronomy. *Astronomy and Astrophysics*, 558. <https://doi.org/10.1051/0004-6361/201322068>
- Bradbury, J., Frostig, R., Hawkins, P., Johnson, M. J., Leary, C., Maclaurin, D., Necula, G., Paszke, A., VanderPlas, J., Wanderman-Milne, S., & Zhang, Q. (2018). *JAX: Composable transformations of Python+NumPy programs* (Version 0.3.13). <http://github.com/google/jax>
- Campagne, J.-E., Lanusse, F., Zuntz, J., Boucaud, A., Casas, S., Karamanis, M., Kirkby, D., Lanzieri, D., Peel, A., & Li, Y. (2023). JAX-COSMO: An End-to-End Differentiable and GPU Accelerated Cosmology Library. *The Open Journal of Astrophysics*, 6, 15. <https://doi.org/10.21105/astro.2302.05163>
- Challinor, A., & Lewis, A. (2011). The linear power spectrum of observed source number counts. *Physical Review D*, 84, 043516. <https://doi.org/10.1103/PhysRevD.84.043516>
- Chisari, N. E., Alonso, D., Krause, E., Leonard, C. D., Bull, P., Neveu, J., Villarreal, A. S., Singh, S., McClintock, T., Ellison, J., Du, Z., Zuntz, J., Mead, A., Joudaki, S., Lorenz, C. S., Tröster, T., Sanchez, J., Lanusse, F., Ishak, M., ... LSST Dark Energy Science Collaboration. (2019). Core Cosmology Library: Precision Cosmological Predictions for

- LSST. *Astrophysical Journal, Supplement*, 242(1), 2. <https://doi.org/10.3847/1538-4365/ab1658>
- Lesgourgues, J. (2011). The Cosmic Linear Anisotropy Solving System (CLASS) I: Overview. *arXiv e-Prints*, arXiv:1104.2932. <https://doi.org/10.48550/arXiv.1104.2932>
- Mattia, A. de. (2022). Cosmoprime: Package for primordial cosmology. In *GitHub repository*. GitHub. <https://github.com/cosmodesi/cosmoprime>
- Planck Collaboration, Aghanim, N., Akrami, Y., Ashdown, M., Aumont, J., Baccigalupi, C., Ballardini, M., Banday, A. J., Barreiro, R. B., Bartolo, N., Basak, S., Battye, R., Benabed, K., Bernard, J.-P., Bersanelli, M., Bielewicz, P., Bock, J. J., Bond, J. R., Borrill, J., ... Zonca, A. (2020). Planck 2018 results - VI. Cosmological parameters. *Astronomy and Astrophysics*, 641, A6. <https://doi.org/10.1051/0004-6361/201833910>
- Rigault, M., Smith, M., Goobar, A., Maguire, K., Dimitriadis, G., Johansson, J., Nordin, J., Burgaz, U., Dhawan, S., Sollerman, J., Regnault, N., Kowalski, M., Nugent, P., Andreoni, I., Amenouche, M., Aubert, M., Barjou-Delayre, C., Bautista, J., Bellm, E., ... Yan, L. (2025). ZTF SN Ia DR2: Overview. *Astronomy and Astrophysics*, 694, A1. <https://doi.org/10.1051/0004-6361/202450388>
- Rubin, D., Aldering, G., Betoule, M., Fruchter, A., Huang, X., Kim, A. G., Lidman, C., Linder, E., Perlmutter, S., Ruiz-Lapuente, P., & Suzuki, N. (2023). Union Through UNITY: Cosmology with 2,000 SNe Using a Unified Bayesian Framework. *arXiv e-Prints*, arXiv:2311.12098. <https://doi.org/10.48550/arXiv.2311.12098>
- Yasuda, N., Tanaka, M., Tominaga, N., Jiang, J., Moriya, T. J., Morokuma, T., Suzuki, N., Takahashi, I., Yamaguchi, M. S., Maeda, K., Sako, M., Ikeda, S., Kimura, A., Morii, M., Ueda, N., Yoshida, N., Lee, C.-H., Suyu, S. H., Komiyama, Y., ... Rubin, D. (2019). The Hyper Suprime-Cam SSP transient survey in COSMOS: Overview. *Publications of the ASJ*, 71(4), 74. <https://doi.org/10.1093/pasj/psz050>